Figure 10.19. The MiniGrid-DoorKey environment. In this environment, the agent (the triangle) must first navigate to the key, pick it up, navigate to the door (the hollow square), open it, and then navigate to the solid square. Each game initializes the objects on the grid randomly, and the agent only has a partial view of the grid indicated by the highlighted region around it.

Figure 10.20. The compatibility function computes the similarity between each key and query vector, resulting in an adjacency matrix.

Figure 10.21. Additive attention is an alternative to dot product attention that can be more stable. Instead of multiplying the queries and keys together, we first pass them independently through linear layers and then add them together, apply a nonlinear function, and pass through another linear layer to change the dimensionality.

Figure 10.22. Top: Log-loss plot during training. The loss drops quickly in the beginning, increases a bit, and then very slowly begins decreasing again. Bottom: Average episode length. This gives us a better idea of the performance of the agent since we can clearly see it is solving the episodes in a shorter number of steps during training.

Figure 10.23. Left: The full observation of the environment. Right: The corresponding partial state view that the agent has access to.

Figure 10.24. Each row corresponds to an attention head (e.g., row 1 corresponds to attention head 1). Left column: The self-attention weights for the agent, which shows the objects to which the agent is paying most attention. Middle column: The self-attention weights for the key, which shows the objects to which the key is paying most attention. Right column: The self-attention weights for the door.

Figure 10.25. Left: Preprocessed state given to the DQN. Middle: Raw video frame. Right: Attention map over state. We can see that the attention map is focused on the alien in the bottom center of the screen, the player in the center, and the bonus at the top, which are the most salient objects in the game.

Appendix Appendix. Mathematics, deep learning, PyTorch